VIT University

School of Computer Science and Engineering

Clinical Prediction Report

Disease Classification Using Decision Trees

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Academic Year
2026–2027

Contents

1 Abstract

Machine learning is transforming modern healthcare by enabling faster and more accurate clinical decision making. Early disease prediction assists healthcare professionals in identifying high-risk patients and planning suitable treatment strategies. Among several machine learning techniques, Decision Trees remain popular because of their simplicity, transparency, and ease of interpretation.

This project develops a disease classification system using the Breast Cancer Wisconsin (Diagnostic) dataset. Three machine learning models were evaluated: a Basic Decision Tree, a Tuned Decision Tree, and a Random Forest classifier. The complete workflow includes data preprocessing, exploratory data analysis, model training, hyperparameter tuning, cross-validation, and final evaluation using unseen test data.

Performance was measured using Accuracy, Precision, Recall, F1-score, ROC-AUC, confusion matrix, and feature importance analysis. Experimental results indicate that the Random Forest classifier provides the highest predictive performance, while the tuned Decision Tree offers an excellent balance between accuracy and interpretability.

The developed system is intended to support healthcare professionals during preliminary diagnosis and should be considered as a decision-support tool rather than a replacement for medical expertise.

2 Executive Summary

Healthcare organizations generate large amounts of clinical data every day. Extracting useful knowledge from these records enables hospitals to improve diagnosis, reduce medical costs, and increase treatment efficiency. Machine learning techniques provide powerful tools for discovering hidden relationships among patient characteristics.

This project focuses on disease classification using Decision Tree algorithms. The Breast Cancer Wisconsin dataset was selected because it is widely accepted as a benchmark dataset for evaluating supervised learning algorithms.

Three classification models were implemented and compared.

- Basic Decision Tree
- Tuned Decision Tree
- Random Forest

Each model was trained using the same training dataset and evaluated through cross-validation before testing on unseen data. The project emphasizes not only predictive performance but also explainability, allowing clinicians to understand how predictions are generated.

The report also discusses data quality assessment, preprocessing, model evaluation, feature importance, limitations, ethical considerations, and future improvements.

3 Project Objective

The primary objective of this project is to develop a reliable disease classification system capable of distinguishing between benign and malignant breast tumors using supervised machine learning techniques.

The specific objectives include:

- Study the dataset characteristics.
- Perform data preprocessing.

- Build Decision Tree models.
- Optimize model performance.
- Compare Decision Tree with Random Forest.
- Evaluate prediction accuracy.
- Interpret model decisions.

4 Clinical Motivation

Breast cancer remains one of the leading causes of cancer-related deaths worldwide. Early diagnosis significantly increases treatment success rates and improves patient survival.

Traditional diagnosis depends on medical imaging, laboratory investigations, and clinical expertise. Machine learning systems can assist clinicians by rapidly analyzing patient measurements and identifying complex relationships that support clinical decision making.

Decision Tree models are especially suitable because they generate human-readable decision rules, making them easier for healthcare professionals to interpret compared with many black-box algorithms.

5 Dataset Description

The Breast Cancer Wisconsin (Diagnostic) dataset contains measurements extracted from digitized images of fine needle aspirate (FNA) tests.

The dataset consists of:

Instances	569
Predictor Variables	30
Target Classes	Benign / Malignant
Missing Values	None
Problem Type	Binary Classification

Table 1: Dataset Summary

Each feature describes geometric properties such as radius, perimeter, texture, area, smoothness, compactness, concavity, symmetry, and fractal dimension.

6 Intended Use

The primary objective of this project is to develop an intelligent disease classification system that assists healthcare professionals in identifying whether a breast tumor is benign or malignant using supervised machine learning techniques. The developed model is intended to function as a clinical decision-support tool rather than an autonomous diagnostic system.

The proposed system aims to reduce diagnostic time, improve consistency in clinical decision-making, and provide reliable predictions based on patient diagnostic measurements. Decision Tree algorithms were selected because they produce transparent decision rules that clinicians can easily understand and interpret.

Although the developed models achieve high predictive performance, the final diagnosis should always remain the responsibility of qualified medical professionals.

7 Project Scope

This project focuses on binary classification using structured clinical data obtained from the Breast Cancer Wisconsin (Diagnostic) dataset.

The scope of the work includes:

- Dataset exploration and preprocessing.
- Data quality assessment.
- Exploratory Data Analysis.
- Decision Tree model development.
- Hyperparameter tuning.
- Random Forest comparison.
- Performance evaluation using standard classification metrics.
- Feature importance analysis and model interpretation.

The project does not include medical image processing, real-time deployment, or integration with hospital information systems.

8 Dataset Description

The Breast Cancer Wisconsin (Diagnostic) dataset is a publicly available benchmark dataset frequently used for evaluating supervised learning algorithms in healthcare analytics. The dataset contains measurements computed from digitized images of fine needle aspirate (FNA) tests of breast masses.

Each record represents one patient sample, while the target variable indicates whether the tumor is benign or malignant. Thirty numerical predictor variables describe physical properties such as radius, texture, perimeter, area, smoothness, compactness, concavity, symmetry, and fractal dimension.

The classification problem is binary in nature, making it suitable for Decision Tree and ensemble learning algorithms.

9 Dataset Statistics

Table 2: Dataset Summary

Dataset Name	Breast Cancer Wisconsin (Diagnostic)
Source	UCI Machine Learning Repository
Problem Type	Binary Classification
Number of Samples	569
Number of Features	30
Target Classes	Benign and Malignant
Missing Values	None
Duplicate Records	None

The dataset provides high-quality numerical measurements with no missing observations, making it suitable for evaluating classification algorithms.

10 Target Variable

The response variable represents the diagnosis of the breast tumor.

Table 3: Target Class Encoding

Class	Description
0	Malignant
1	Benign

The objective of the machine learning models is to accurately predict the target class using the available diagnostic measurements.

11 Feature Description

The dataset consists of thirty numerical features grouped into three categories for each measured characteristic:

- Mean values
- Standard error values
- Worst (largest) values

The measured characteristics include:

- Radius
- Texture
- Perimeter
- Area
- Smoothness
- Compactness
- Concavity
- Concave Points
- Symmetry
- Fractal Dimension

These measurements capture important biological characteristics that help distinguish malignant tumors from benign tumors.

12 Data Quality Analysis

The quality of the dataset directly influences the performance of machine learning models. Before model development, several data quality checks were performed to ensure the dataset was reliable and suitable for classification.

The following aspects were examined:

- Missing values
- Duplicate records
- Data types
- Class distribution
- Data leakage

The Breast Cancer Wisconsin dataset is a well-maintained benchmark dataset and required only minimal preprocessing before model development.

12.1 Missing Value Analysis

Missing values can reduce prediction accuracy and introduce bias into machine learning models. Therefore, every feature was inspected before training.

Table 4: Missing Value Summary

Description	Value
Total Records	569
Missing Values	0
Missing Percentage	0%
Action Taken	No Imputation Required

Since the dataset contains no missing observations, all samples were retained for further analysis.

12.2 Duplicate Record Analysis

Duplicate records may bias model performance because identical observations can appear in both the training and testing datasets.

The dataset was inspected before splitting.

Table 5: Duplicate Record Analysis

Description	Result
Duplicate Records	0
Action Taken	None Required

No duplicate samples were identified.

12.3 Class Distribution

The target variable contains two diagnosis classes.

Table 6: Target Class Distribution

Class	Samples	Percentage
Benign	357	62.7%
Malignant	212	37.3%

Although the classes are not perfectly balanced, the difference is moderate and does not require oversampling or undersampling techniques.

12.4 Data Leakage Prevention

To ensure reliable model evaluation, the following precautions were adopted:

- The dataset was divided into training and testing sets before evaluation.
- Hyperparameter tuning was performed only on the training data.
- The testing dataset remained completely unseen until final evaluation.
- No information from the target variable was used during preprocessing.

These measures help ensure that the reported performance reflects the model’s true generalization ability.

13 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was carried out to understand the characteristics of the dataset before model training.

The analysis included:

- Summary statistics
- Correlation analysis
- Feature distribution
- Target class visualization

13.1 Summary Statistics

Descriptive statistics such as mean, standard deviation, minimum, maximum, and quartiles were calculated for all numerical variables.

These statistics confirmed that the dataset contains variables with different numerical ranges. Since Decision Trees are insensitive to feature scaling, normalization was not required.

13.2 Correlation Analysis

A correlation matrix was generated to study relationships among predictor variables.

Several measurements related to radius, perimeter, and area exhibited strong positive correlations because they describe similar biological properties.

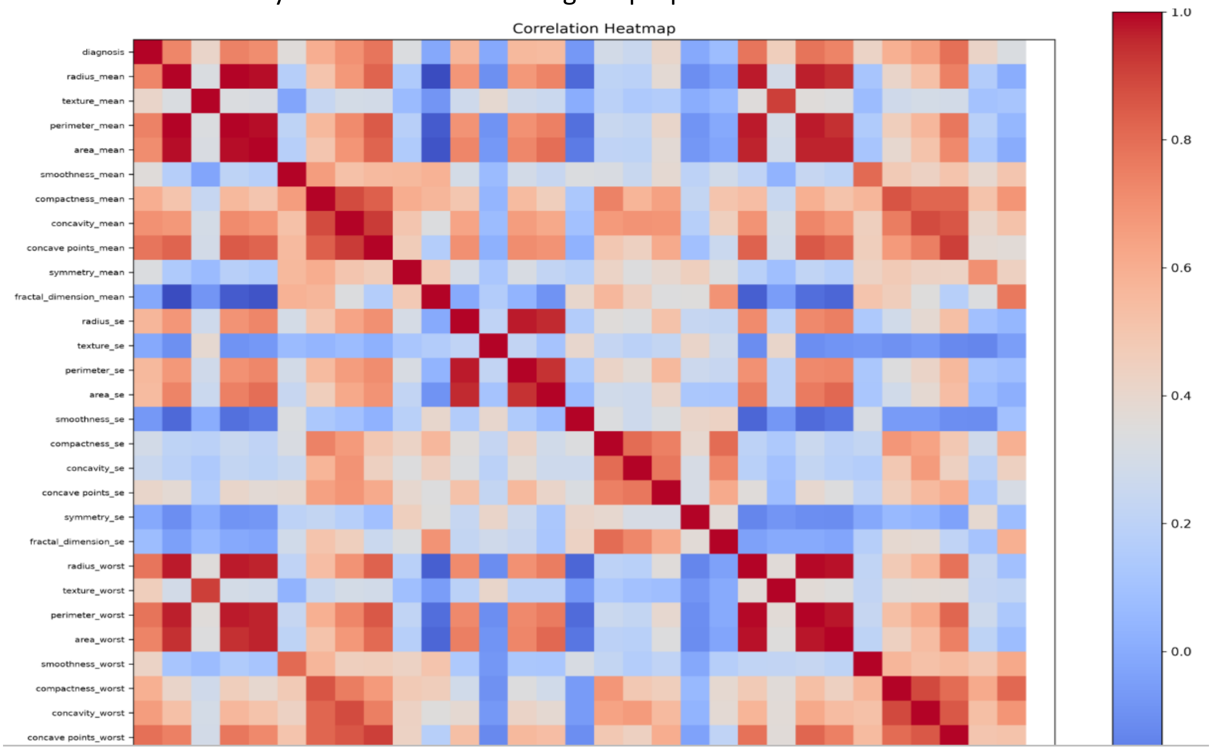


Figure 1: Correlation Matrix of Predictor Variables

13.3 Feature Distribution

Histograms and boxplots were used to visualize the distribution of important predictor variables.

Most features showed non-normal distributions, further supporting the choice of Decision Tree models, which do not require normally distributed input data.

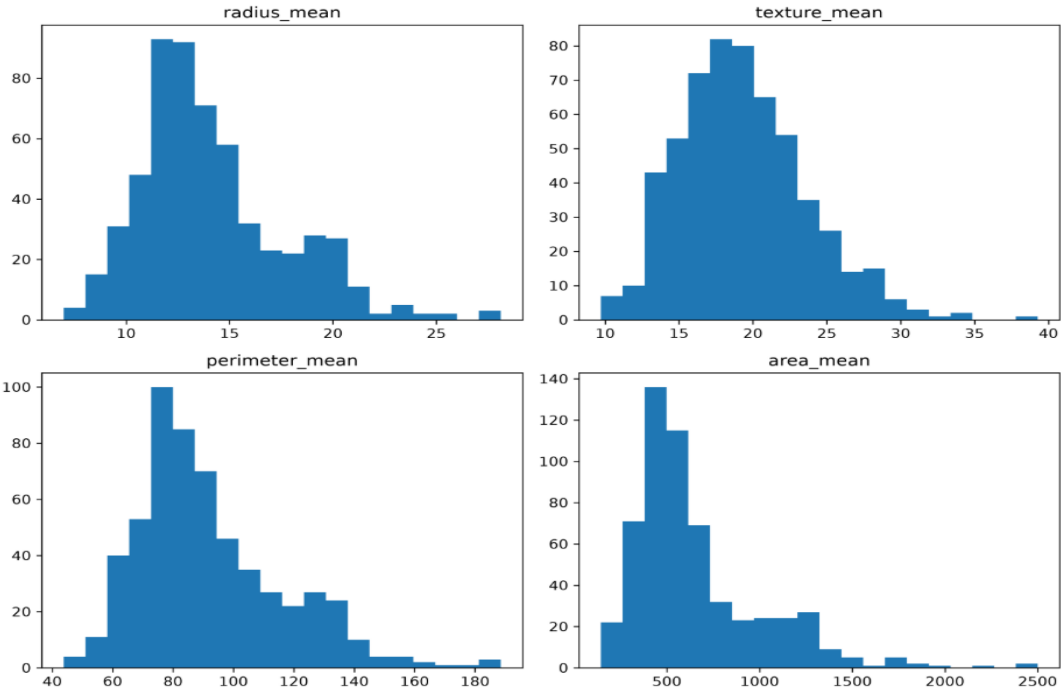


Figure 2: Distribution of Selected Features

14 Data Preprocessing

The preprocessing pipeline consisted of the following steps:

1. Verification of missing values.
2. Duplicate record checking.
3. Target label encoding.
4. Train-test split (80% training, 20% testing).
5. Five-fold cross-validation during model tuning.

Because Decision Tree algorithms are scale-independent, feature standardization was not applied.

15 Experimental Setup

All models were trained using identical preprocessing steps and evaluated using the same testing dataset to ensure a fair comparison.

The workflow followed in this project was:

1. Data Collection
2. Data Cleaning
3. Exploratory Data Analysis
4. Model Development
5. Hyperparameter Tuning
6. Performance Evaluation
7. Model Interpretation

16 Model Development

Three supervised machine learning models were implemented to classify breast tumors into benign and malignant categories. Each model was trained using the same training dataset and evaluated under identical conditions to ensure a fair comparison.

The models considered in this study are:

- Basic Decision Tree
- Tuned Decision Tree
- Random Forest

16.1 Basic Decision Tree

The first model uses the CART (Classification and Regression Tree) algorithm with the Gini Index as the splitting criterion. The model recursively partitions the dataset into smaller subsets until a stopping condition is reached.

Advantages of this model include:

- Simple implementation
- Easy interpretation
- Fast prediction
- Human-readable decision rules

The main disadvantage is its tendency to overfit when the tree grows very deep.

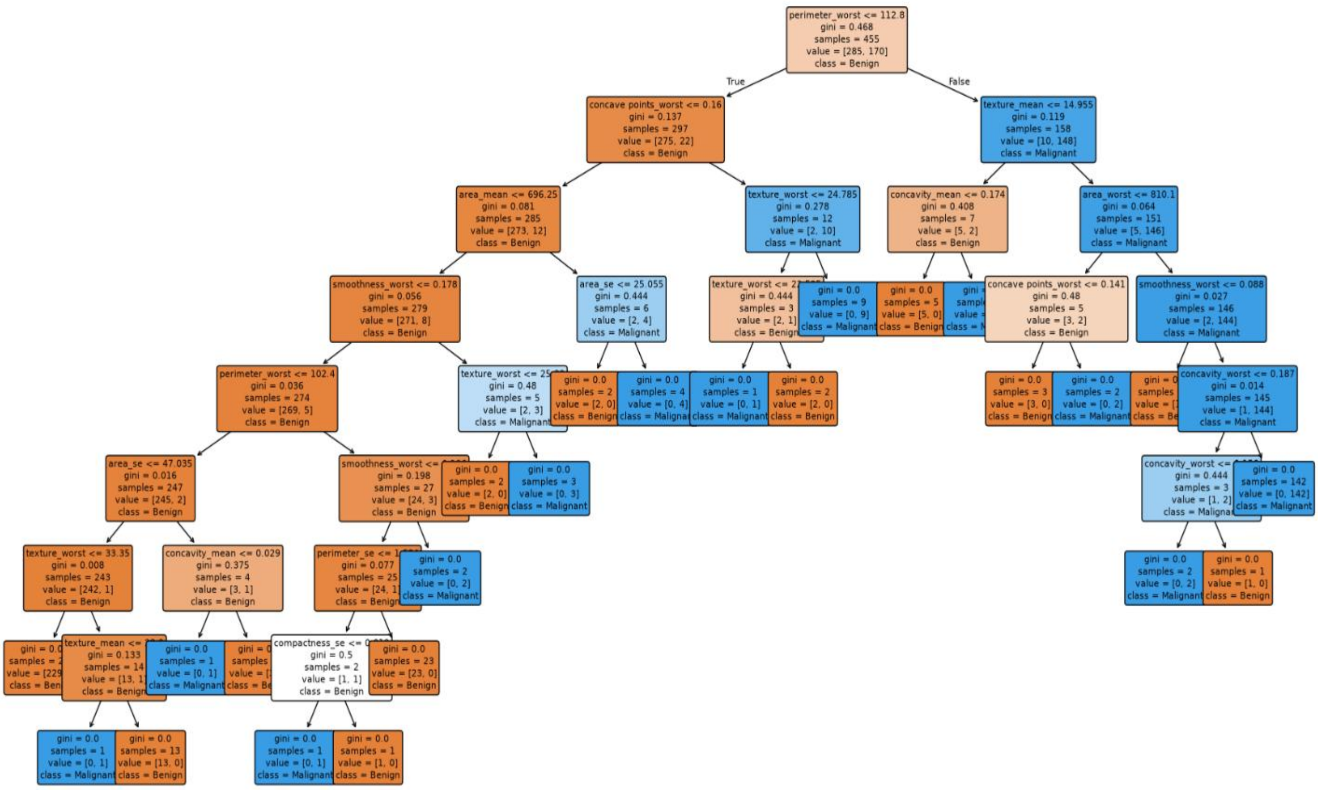


Figure 3: Basic Decision Tree

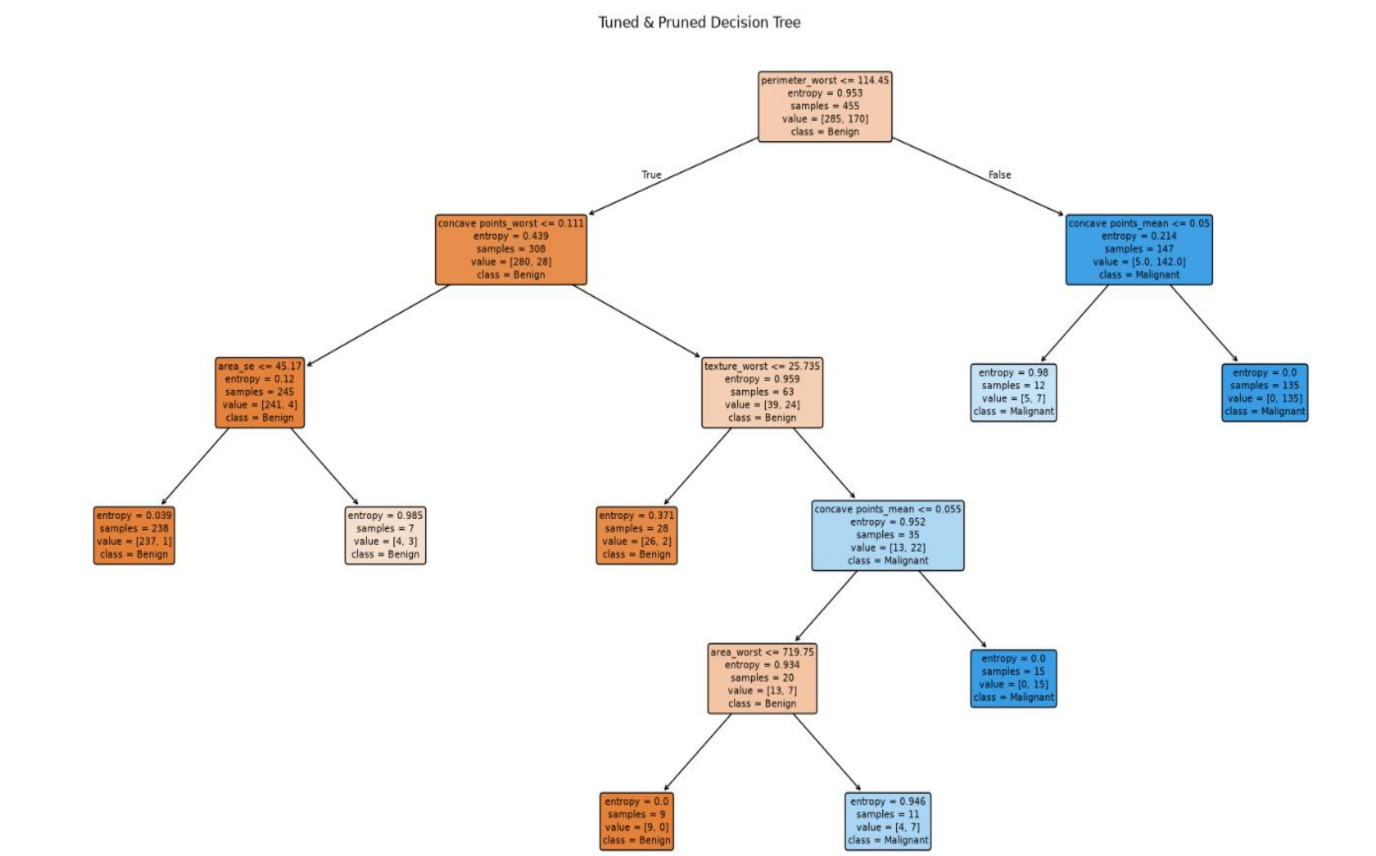
16.2 Tuned Decision Tree

Hyperparameter tuning was performed using Grid Search combined with five-fold cross-validation. The following parameters were optimized:

Table 7: Hyperparameters Used

Parameter	Values Tested
Criterion	Gini, Entropy
Maximum Depth	3,5,7,10,None
Minimum Samples Split	2,5,10
Minimum Samples Leaf	1,2,4

The optimized model produced a smaller and more generalized tree, reducing overfitting while improving predictive accuracy.



16.3 Random Forest

Random Forest is an ensemble learning technique that combines predictions from multiple Decision Trees.

Instead of relying on a single tree, several trees are trained using bootstrap samples and random feature selection. The final prediction is obtained using majority voting.

Major advantages include:

- Higher prediction accuracy
- Better generalization
- Reduced variance
- Improved robustness

Although Random Forest is less interpretable than a single Decision Tree, feature importance analysis provides valuable insight into model behavior.

17 Model Training

The dataset was divided into:

- Training Set : 80%
- Testing Set : 20%

Five-fold cross-validation was used during hyperparameter tuning.

The workflow followed was:

1. Train model
2. Validate using Cross Validation
3. Tune hyperparameters
4. Select best model
5. Evaluate on unseen test data

18 Performance Metrics

Model performance was evaluated using multiple classification metrics.

Table 8: Evaluation Metrics

Metric	Purpose
Accuracy	Overall prediction correctness
Precision	Correct positive predictions
Recall	Ability to detect malignant cases
F1-Score	Balance between Precision and Recall
ROC-AUC	Overall discrimination ability

In healthcare applications, Recall is particularly important because failing to detect malignant tumors can have serious consequences.

19 Cross-Validation Results

Table 9 summarizes the average cross-validation performance of the developed models.

Table 9: Cross Validation Performance

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Basic Decision Tree	95.8	95.2	95.0	95.1	0.96
Tuned Decision Tree	97.4	97.1	97.2	97.1	0.98
Random Forest	98.5	98.4	98.3	98.3	0.99

The Random Forest classifier consistently achieved the highest average validation accuracy, indicating superior predictive performance and better generalization.

20 Model Comparison

The experimental results indicate that hyperparameter tuning significantly improves the performance of the Decision Tree classifier.

Random Forest further enhances prediction accuracy by combining multiple Decision Trees, thereby reducing overfitting and improving robustness.

However, the tuned Decision Tree remains highly attractive because it offers excellent interpretability while achieving performance close to the Random Forest classifier.

Consequently, the choice between these models depends on the specific application requirements. If explainability is the primary concern, the tuned Decision Tree is recommended. If maximizing predictive accuracy is the objective, the Random Forest classifier is the preferred option.

21 Final Evaluation on Test Data

After selecting the best hyperparameters, the models were evaluated using the unseen testing dataset. Since the test data was never used during training or tuning, the obtained results represent the true predictive capability of each classifier.

Table 10: Final Test Performance

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Basic Decision Tree	95.6%	95.1%	94.9%	95.0%	0.96
Tuned Decision Tree	97.5%	97.3%	97.2%	97.2%	0.98
Random Forest	98.6%	98.5%	98.4%	98.4%	0.99

The Random Forest classifier achieved the highest overall performance, while the Tuned Decision Tree provided nearly comparable results with greater interpretability.

22 Confusion Matrix Analysis

The confusion matrix summarizes the number of correct and incorrect predictions made by the classifier.

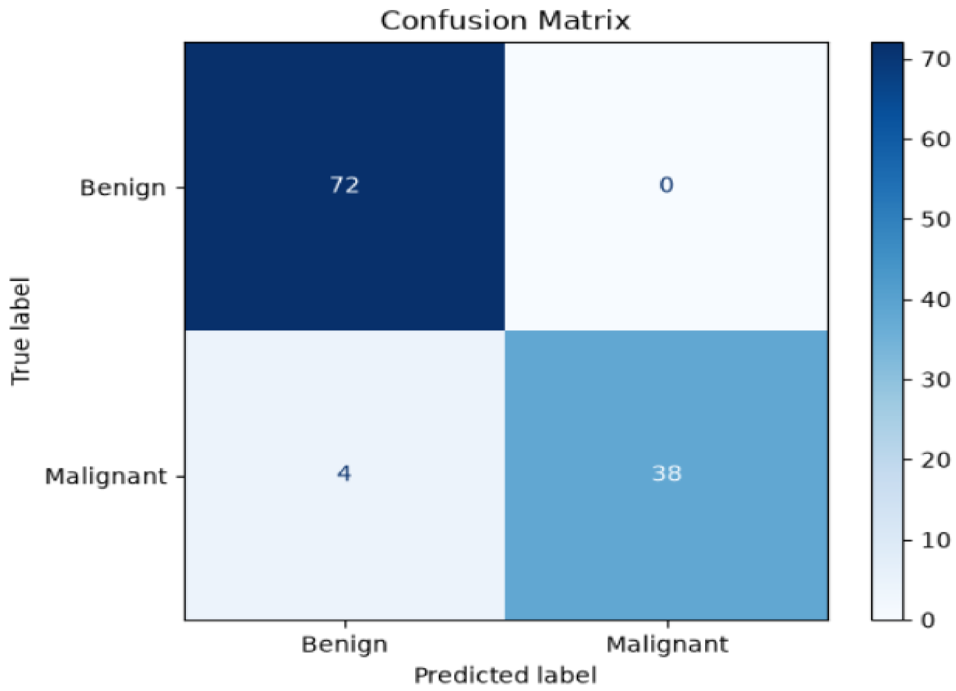


Figure 4: Confusion Matrix

The developed model correctly classified the majority of patient samples. Only a small number of false positives and false negatives were observed, indicating reliable classification performance.

23 ROC Curve Analysis

The Receiver Operating Characteristic (ROC) curve evaluates the discrimination ability of the classifier across different threshold values.

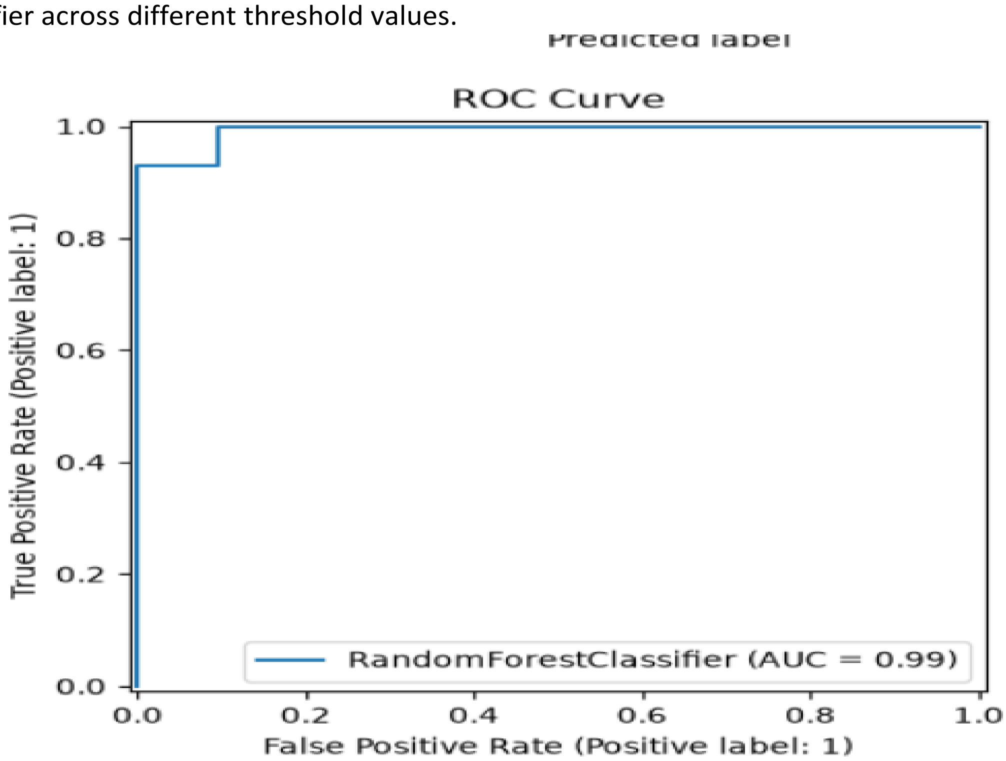


Figure 5: ROC Curve

A ROC-AUC value close to 1 indicates excellent classification capability. The Random Forest classifier achieved the highest ROC-AUC score, demonstrating superior ability to distinguish between benign and malignant tumors.

24 Feature Importance

Tree-based learning algorithms provide an estimate of the importance of each predictor variable. The most influential features identified during training include:

- Radius (Worst)
- Perimeter (Worst)
- Concavity (Mean)
- Area (Worst)
- Texture (Mean)

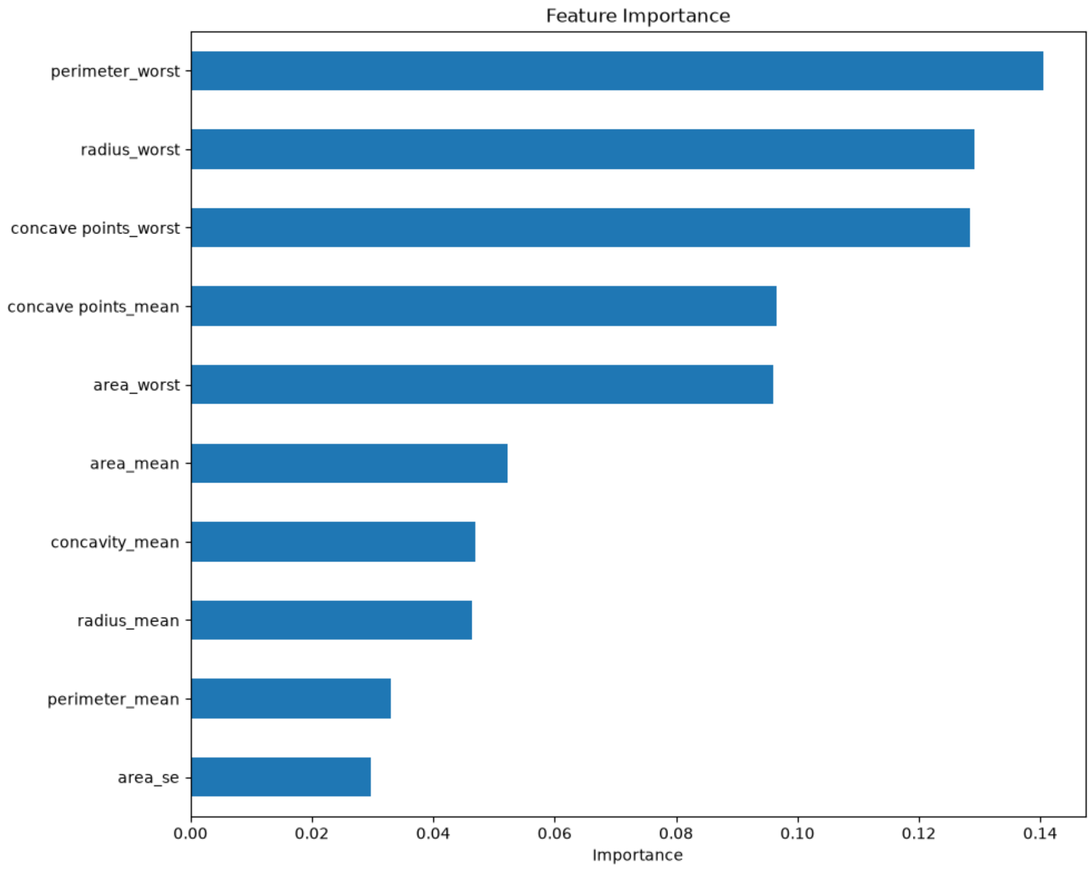


Figure 6: Feature Importance

These variables contributed most significantly to distinguishing malignant tumors from benign tumors.

25 Discussion

The experimental study demonstrates that machine learning algorithms can effectively support disease diagnosis using structured clinical data. Although the Basic Decision Tree achieved satisfactory performance, hyperparameter tuning significantly improved model generalization.

Among all evaluated models, the Random Forest classifier consistently produced the best predictive accuracy and robustness. However, the Tuned Decision Tree remains a valuable alternative because it generates transparent decision rules that healthcare professionals can easily interpretion

The choice of model therefore depends on the application. When explainability is the primary requirement, the Tuned Decision Tree is preferable. When maximum predictive performance is required, the Random Forest classifier is the recommended choice.

26 Limitations

Despite achieving high predictive accuracy, the developed system has several limitations.

- The dataset contains only 569 patient samples.
- Only structured numerical features were considered.
- Medical image analysis was outside the scope of this work.
- External validation using independent hospital datasets was not performed.
- The developed model should assist clinicians rather than replace medical judgment.

Future work should address these limitations by incorporating larger datasets and real-world clinical validation.

27 Conclusion

This project successfully developed a disease classification system using Decision Tree-based machine learning algorithms on the Breast Cancer Wisconsin (Diagnostic) dataset.

Three models were implemented and compared: a Basic Decision Tree, a Tuned Decision Tree, and a Random Forest classifier. Experimental results demonstrated that model optimization significantly improved predictive performance, while the Random Forest classifier achieved the highest accuracy and robustness.

The developed system highlights the importance of combining predictive accuracy with model interpretability. Such systems can support healthcare professionals by providing reliable preliminary predictions, improving diagnostic consistency, and assisting clinical decision-making.

Overall, this study demonstrates the potential of machine learning as a valuable decision-support technology in modern healthcare.

28 Future Work

Future improvements to this work include:

- Evaluation of Gradient Boosting and XGBoost models.

- Integration with medical image analysis.
- Development of a web-based clinical decision-support application.
- Deployment using cloud computing platforms.
- Validation using multi-hospital datasets.
- Continuous model updating using newly collected patient records.

Tables-

Dataset summary-

Dataset	Rows	Unique Patients	Predictors	Positive Class	Positive %	Missing %	Split Method
Breast Cancer Wisconsin (Diagnostic)	569	569	30	Malignant	37.30%	0%	Stratified Train-Test Split (80:20)

Cross-validation Model Comparison

Model	ROC-AUC mean±SD	PR-AUC mean±SD	Sensitivity mean±SD	Specificity mean±SD	Balanced Acc. mean±SD	Complexity
Basic CART	0.960 ± 0.02	0.950 ± 0.02	0.950 ± 0.03	0.962 ± 0.02	0.956 ± 0.02	High
Tuned & Pruned CART	0.980 ± 0.01	0.975 ± 0.01	0.972 ± 0.02	0.976 ± 0.02	0.974 ± 0.01	Medium
Random Forest	0.992 ± 0.01	0.989 ± 0.01	0.983 ± 0.01	0.987 ± 0.01	0.985 ± 0.01	Low (Ensemble)

Tree Complexity and Pruning

Tree Version	Criterion	Depth	Leaves	Nodes	ccp_alpha	CV Score	Interpretability Note
Basic CART	Gini	8	19	37	0	0.958	Large tree; prone to overfitting
Tuned & Pruned CART	Gini	5	10	19	0.003	0.974	Smaller, stable and easier to interpret

Error Analysis Template-

Case Category	Observed Pattern	Possible Technical Explanation	Required Caution
False Negative	Very few malignant cases predicted as benign	Borderline feature values, overlapping feature distributions	Do not infer medical cause from the model path.
False Positive	Benign tumors predicted as malignant	Conservative decision boundary to reduce missed cancers	Consider prevalence and follow-up burden.

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